Akoizo Original Practice - Form 1

Division C | Wright Stuff | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) What is the minimum airplane mass excluding the motor?

Unit: Rules, scoring, and the flight log bonus.
Worked-case data: A team with a complete flight log flies official flights of 96.4 s and 112.0 s. What is its score and tiebreaker?
Separate investigation observations:
Team A: Best flight 120 s, no flight log.
Team B: Best flight 105 s, complete flight log.
Team C: Best flight 112 s, partial flight log.

A. 8.0 g

B. 1.5 g

C. 3.5 g

D. 10.0 g

Answer: _____

2. (2 points) Why is a stiff motor stick important?

Unit: Building light and strong.
Worked-case data: A built airframe weighs 7.4 g. The CG is slightly behind the target. How should the team meet the 8.0 g minimum?
Separate investigation observations:
Flight 1: Fresh build, 102 s.
Flight 6: After a week in a hot car, 61 s; plane turned sharply right.
Inspection: Right wing twisted 2 degrees.

A. It reduces motor mass

B. It shortens the flight

C. Bending changes thrust direction and trim as torque changes

D. It adds lift

Answer: _____

3. (2 points) Why are large, slow propellers efficient for indoor planes?

Unit: Propellers: turning torque into thrust.
Worked-case data: A plane cruises at 1.2 m/s and its propeller effectively advances 0.24 m per turn. A motor has 1,200 usable turns. Estimate motor run time.
Separate investigation observations:
Prop 1: Diameter 22 cm; plane cruised level; motor ran 140 s.
Prop 2: Diameter 26 cm, same pitch; plane sank after the burst; 90 s.
Prop 3: Diameter 18 cm; plane climbed fast; motor ran 95 s.

A. They make more noise

B. They spin faster

C. They push a large mass of air gently, wasting less energy

D. They weigh less

Answer: _____

4. (2 points) Which drag type is largest at low speed?

Unit: Wings and slow flight.
Worked-case data: A 9.5 g plane (with motor) has 0.040 m^2 of wing and C_L = 0.8. Find its level-flight speed (ρ = 1.2 kg/m^3).
Separate investigation observations:
Plane 1: Wing area 0.030 m^2; cruise 2.6 m/s; flight 85 s.
Plane 2: Wing area 0.042 m^2; cruise 2.1 m/s; flight 118 s.
Same: Mass, motor, and turns.

A. Skin friction only

B. Wave drag

C. Induced drag

D. Parasite drag

Answer: _____

5. (2 points) Why bank a safe official flight early?

Unit: Testing, the flight log, and competition day.
Worked-case data: Log data at a 9 m ceiling: torque 0.10 -> 95 s, 0.13 -> 118 s, 0.16 -> 104 s (ceiling bumps). Plan the first official flight.
Separate investigation observations:
Room A, 7 m ceiling: Best time at torque 0.10.
Room B, 11 m ceiling: Best time at torque 0.15.
Competition: Ceiling announced as 9 m.

A. It guarantees a score before trying a riskier winding

B. Official flights get bonus time

C. The first flight counts double

D. It is required

Answer: _____

6. (4 points) Why test glide before powered flights?

Unit: Stability, trim, and circling flight.
Worked-case data: A plane stalls on every glide test. List adjustments in order to try.
Separate investigation observations:
Glide: Smooth and straight.
Powered launch: Plane rolls left and spirals down in the first 5 s.
After 5 s: Torque drops; plane recovers if still airborne.

Answer: _____
_____
_____

7. (4 points) Why is the motor mass limit the main energy limit?

Unit: Rubber motors: storing and releasing energy.
Worked-case data: A team uses a 15:1 winder and tests show a motor breaks near 1,400 turns. How many cranks should they wind for a safe 8 percent back-off?
Separate investigation observations:
Motor A: 1.4 g, 30 cm long; 1,100 turns; climbed to ceiling, 78 s.
Motor B: 1.4 g, 38 cm long; 1,350 turns; stayed below ceiling, 104 s.
Ceiling: 8 m in both tests.

Answer: _____
_____
_____

8. (4 points) Why does the 8.0 g minimum change design priorities?

Answer: _____
_____
_____

9. (4 points) Why test box fit with the exact flight setup?

Unit: Building light and strong.
Worked-case data: A built airframe weighs 7.4 g. The CG is slightly behind the target. How should the team meet the 8.0 g minimum?
Separate investigation observations:
Flight 1: Fresh build, 102 s.
Flight 6: After a week in a hot car, 61 s; plane turned sharply right.
Inspection: Right wing twisted 2 degrees.

Answer: _____
_____
_____

10. (4 points) Why is a freewheel allowed but variable pitch not?

Unit: Propellers: turning torque into thrust.
Worked-case data: A plane cruises at 1.2 m/s and its propeller effectively advances 0.24 m per turn. A motor has 1,200 usable turns. Estimate motor run time.
Separate investigation observations:
Prop 1: Diameter 22 cm; plane cruised level; motor ran 140 s.
Prop 2: Diameter 26 cm, same pitch; plane sank after the burst; 90 s.
Prop 3: Diameter 18 cm; plane climbed fast; motor ran 95 s.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A 9.5 g plane (with motor) has 0.040 m^2 of wing and C_L = 0.8. Find its level-flight speed (ρ = 1.2 kg/m^3).

Answer: _____
_____
_____

12. (6 points) Solve this case and explain your reasoning: Log data at a 9 m ceiling: torque 0.10 -> 95 s, 0.13 -> 118 s, 0.16 -> 104 s (ceiling bumps). Plan the first official flight.

Answer: _____
_____
_____

13. (2 points) What is the likely cause?

Supplied synthetic observations:
Glide: Smooth and straight.
Powered launch: Plane rolls left and spirals down in the first 5 s.
After 5 s: Torque drops; plane recovers if still airborne.

A. Torque roll during the burst is too strong for the trim

B. CG too far forward

C. Not enough motor turns

Answer: _____

14. (2 points) Define “Rubber motor” in the context of Wright Stuff.

Answer: _____
_____
_____

15. (2 points) Define “Airframe mass” in the context of Wright Stuff.

Answer: _____
_____
_____

16. (2 points) Define “Rib” in the context of Wright Stuff.

Answer: _____
_____
_____

17. (2 points) Define “Pitch” in the context of Wright Stuff.

Answer: _____
_____
_____

18. (2 points) Define “Camber” in the context of Wright Stuff.

Answer: _____
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